

Day 10 — Sigma CLI: Converting Rules to Production SIEM Queries

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Objective

To install sigma-cli and its backend plugins, convert all 3 Day 9 Sigma rules into Splunk SPL, Elastic Lucene, and QRadar AQL, validate each conversion against live Splunk and Kibana instances, and deploy 5 SigmaHQ community rules into Wazuh as active detections.

Part 1 — Installing sigma-cli and Backend Plugins

Install sigma-cli:

- `pip install sigma-cli`

Add to PATH (installed to user-local bin, not on PATH by default):

- `echo 'export PATH="$HOME/.local/bin:$PATH"' >> ~/.bashrc`
`source ~/.bashrc`

Verify installation:

- `sigma --help`

Install the 3 backend plugins:

- `pip install pysigma-backend-splunk`
- `pip install pysigma-backend-elasticsearch`
- `pip install pysigma-backend-qradar`

Note: Installing the QRadar backend downgraded pySigma to an incompatible version, breaking sigma-cli and the other two backends. Fixed by reinstalling the correct version:

- `pip install "pysigma>=1.3.0,<2.0.0" --upgrade`

Confirm all 3 backends registered correctly:

- sigma list targets

```
fara@ubuntu:~$ sigma list targets
```

Identifier	Target Query Language	Processing Pipeline Required	Plugin	Processor
lucene	Elasticsearch Lucene		elasticsearch	Yes
eql	Elasticsearch EQL		elasticsearch	Yes
esql	ES QL backend		elasticsearch	Yes
elastalert	Elasticsearch Elastalert		elasticsearch	Yes
qradar	Base backend		qradar	No
splunk	Splunk SPL & tstats data model queries		splunk	Yes
splunk_spl2	Splunk SPL2 queries		splunk	Yes

Identify correct processing pipelines for Windows logs:

bash

- sigma list pipelines

Result: splunk_windows (Splunk), ecs_windows (Elastic), qradar_windows (QRadar)

Part 2 — Deliverable 1

Converting the 3 Day 9 Rules to All 3 SIEM Formats

Rule 1 — Windows Defender Disabled:

```
sigma convert -t splunk -p splunk_windows ~/sigma-rules/rule1-defender-disabled.yml
```

```
sigma convert -t lucene -p ecs_windows ~/sigma-rules/rule1-defender-disabled.yml
```

```
sigma convert -t qradar -p qradar_windows ~/sigma-rules/rule1-defender-disabled.yml
```

Target	Output
Splunk	EventCode=5001
Elastic (Lucene)	EventID:5001
QRadar	SELECT UTF8(payload) as search_payload from events where "EventID"=5001

```
fara@ubuntuerver:~$ sigma convert -t splunk -p splunk_windows
~/sigma-rules/rule1-defender-disabled.yml
Parsing Sigma rules [_____]
Parsing Sigma rules [#####] 10
0%
EventCode=5001
fara@ubuntuerver:~$
```

```
fara@ubuntuerver:~$ sigma convert -t lucene -p ecs_windows
~/sigma-rules/rule1-defender-disabled.yml
Parsing Sigma rules [_____]
Parsing Sigma rules [#####]
100%
EventID:5001
fara@ubuntuerver:~$
```

```
fara@ubuntuerver:~$ sigma convert -t qradar -p qradar
_windows ~/sigma-rules/rule1-defender-disabled.yml
Parsing Sigma rules [_____]
Parsing Sigma rules [#####]
####] 100%
SELECT UTF8(payload) as search_payload from events whe
re "EventID"=5001
fara@ubuntuerver:~$
```

Rule 2 — PowerShell Encoded Command:

```
sigma convert -t splunk -p splunk_windows ~/sigma-rules/rule2-powershell-encoded.yml
```

```
sigma convert -t lucene -p ecs_windows ~/sigma-rules/rule2-powershell-encoded.yml
```

```
sigma convert -t qradar ~/sigma-rules/rule2-powershell-encoded.yml
```

Target	Output
Splunk	source IN ("WinEventLog:Microsoft-Windows-PowerShell/Operational", "WinEventLog:PowerShellCore/Operational") EventCode=4104 CommandLine="*-EncodedCommand"
Elastic (Lucene)	(winlog.channel:(Microsoft\Windows\PowerShell\Operational OR PowerShellCore\Operational)) AND (event.code:4104 AND winlog.event_data.CommandLine:*-EncodedCommand*)
QRadar	SELECT UTF8(payload) as search_payload from events where "EventID"=4104 AND "Process CommandLine" ILIKE '%-EncodedCommand%' (required --without-pipeline, since qradar_windows pipeline doesn't support the powershell service)

```
fara@ubuntu:~$ sigma convert -t splunk -p splunk_windows ~/sigma-rules/rule2-powershell-encoded.yml
sigma convert -t lucene -p ecs_windows ~/sigma-rules/rule2-powershell-encoded.yml
sigma convert -t qradar -p qradar_windows ~/sigma-rules/rule2-powershell-encoded.yml
Parsing Sigma rules [#####] 100%
source IN ("WinEventLog:Microsoft-Windows-PowerShell/Operational", "WinEventLog:PowerShellCore/Operational")
EventCode=4104 CommandLine="*-EncodedCommand"
Parsing Sigma rules [#####] 100%
(winlog.channel:(Microsoft\Windows\PowerShell\Operational OR PowerShellCore\Operational)) AND (event.code
:4104 AND winlog.event_data.CommandLine:*-EncodedCommand*)
```

```
fara@ubuntu:~$ sigma convert -t qradar ~/sigma-rules/rule2-powershell-encoded.yml
Parsing Sigma rules [#####] 100%
SELECT UTF8(payload) as search_payload from events where "EventID"=4104 AND "Process CommandLine"
ILIKE '%-EncodedCommand%'
fara@ubuntu:~$
```

Rule 3 — New Admin Account:

```
sigma convert -t splunk -p splunk_windows ~/sigma-rules/rule3-new-admin-account.yml
```

```
sigma convert -t lucene -p ecs_windows ~/sigma-rules/rule3-new-admin-account.yml
```

```
sigma convert -t qradar ~/sigma-rules/rule3-new-admin-account.yml
```

Target	Output
Splunk	source="WinEventLog:Security" EventCode=4720 OR (EventCode=4732 Group_Name="*Administrators*")
Elastic (Lucene)	winlog.channel:Security AND (event.code:4720 OR (event.code:4732 AND winlog.event_data.Group_Name:*Administrators*))
QRadar	Conversion failed — Error: The QRadar savedsearches Sigma backend supports only the following fields for process_creation log source. Isolated via a stripped-down test rule (single selection, no OR/modifier) that converted successfully, confirming the QRadar backend cannot handle multi-selection rules combined with field modifiers. No workaround found.

```
fara@ubuntuuserver:~$ sigma convert -t splunk -p splunk_windows ~/sigma-rules/rule3-new-admin-account.yml
Parsing Sigma rules [#####] 100%
source="WinEventLog:Security" EventCode=4720 OR (EventCode=4732 Group_Name="*Administrators*")
fara@ubuntuuserver:~$ sigma convert -t lucene -p ecs_windows ~/sigma-rules/rule3-new-admin-account.yml
Parsing Sigma rules [#####] 100%
winlog.channel:Security AND (event.code:4720 OR (event.code:4732 AND winlog.event_data.Group_Name:*Administrators
fara@ubuntuuserver:~$
```

Part 3 — Validating Conversions Against Live SIEMs

Each Splunk query was pasted into Splunk's Search & Reporting app; each Elastic query was pasted into Kibana Discover.

All 6 queries (3 Splunk + 3 Elastic) executed successfully with no syntax errors, returning 0 results — expected, since no live Windows telemetry matching these specific events was present at the time of testing.

splunk> Enterprise

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

New Search

Save As

EventCode=5001

Time range: Last 24 hours

✓ 0 events (10/07/2026 14:43:00.000 to 10/07/2026 15:43:51.000)

No Event Sampling

Events (0) Patterns Statistics Visualization

No results found. Try expanding the time range.

New Search

Save As

source="WinEventLog:Security" EventCode=4720 OR (EventCode=4732 Group_Name="*Administrators*")

Time range: Last 24 hours

✓ 0 events (10/07/2026 14:50:00.000 to 10/07/2026 15:50:02.000)

No Event Sampling

Events (0) Patterns Statistics Visualization

No results found. Try expanding the time range.

Search & Reporting

Search

Analytics

Datasets

Reports

Alerts

Dashboards

New Search

Save As

source IN ("WinEventLog:Microsoft-Windows -PowerShell/Operational", "WinEventLog :PowerShellCore/Operational") EventCode=4104 CommandLine="*-EncodedCommand*"

Time range: Last 24 hours

✓ 0 events (10/07/2026 14:49:00.000 to 10/07/2026 15:49:28.000)

No Event Sampling

Events (0) Patterns Statistics Visualization

No results found. Try expanding the time range.

 Find apps, content, and more. ^/

Discover Check out context-aware Discover ?

Untitled +

Data view logs-generic-default* EventID:5001 Last 15 days

Search field names 0

Available fields 45

- @timestamp
- @version
- agent.ephemeral_id
- agent.id
- agent.name
- agent.type
- agent.version
- bytes
- client_ip
- data_stream.dataset



No results match

Share

(winlog.channel:(Microsoft\Windows\PowerShell\Operational OR PowerShellCore\Operational)) AND (event.code:4104 AND winlog.event_data.CommandLine:*{-EncodedCommand*})

No results match your search criteria



efault* winlog.channel:Security AND (event.code:4720 OR (event.code:4732 AND winlog.event_data.Group_Name:*Administrators*))

45

No results match your search criteria

Part 4 — Deliverable 2

Deploying 5 SigmaHQ Community Rules to Wazuh

Finding: No official Wazuh backend exists for sigma-cli (pysigma-backend-wazuh not found on PyPI), and Wazuh is not listed as a supported platform on Uncoder.IO. Rules were instead converted using **SigWaz** (<https://sigwaz.com/converter>), a community-built Sigma-to-Wazuh XML converter, with manual fixes applied to its output where needed

5 rules selected from:

https://github.com/SigmaHQ/sigma/tree/master/rules/windows/process_creation

#	Rule	Wazuh Rule ID	MITRE Technique
1	Notepad Password Files Discovery	900000	T1083
2	Whoami Utility Execution	900001, 900002	T1033
3	Windows Firewall Disabled via PowerShell	900003	T1685
4	Port Forwarding Activity Via SSH.EXE	900004	T1572
5	Delete All Scheduled Tasks	900005	T1489

Deployment process (repeated per rule):

- `cat > ~/community-ruleX.xml << 'EOF'`
[XML content]
`EOF`
- `docker cp ~/community-ruleX.xml single-node-wazuh.manager-1:/var/ossec/etc/rules/community_ruleX.xml`
- `docker exec single-node-wazuh.manager-1 /var/ossec/bin/wazuh-control start`

Each rule verified via Wazuh Dashboard → Management → Rules → searched by rule ID.

Key technical finding: Rules using type="pcre2" correctly handle escaped square brackets ([,]) in regex, resolving the exact bracket-escaping limitation encountered with Wazuh's default regex engine on Day 7.

Rules (6) [Manage rules files](#) [Add new rules file](#) [Refresh](#) [Export form](#)

From here you can manage your rules.

90000					WQL	Custom r
ID ↑	Description	Groups	Regulatory compliance	Level	File	Path
900000	Notepad Password Files Discovery	windows, process_creation, sigma	MITRE	7	community_rule1.xml	etc/rules
900001	Whoami Utility Execution	windows, process_creation, sigma	MITRE	7	community_rule2.xml	etc/rules
900002	Whoami Utility Execution (by original filename)	windows, process_creation, sigma	MITRE	7	community_rule2.xml	etc/rules
900003	Windows Firewall Disabled via PowerShell	windows, process_creation, sigma	MITRE	10	community_rule3.xml	etc/rules
900004	Port Forwarding Activity Via SSH.EXE	windows, process_creation, sigma	MITRE	10	community_rule4.xml	etc/rules
900005	Delete All Scheduled Tasks	windows, process_creation,	MITRE	12	community_rule5.xml	etc/rules

Conclusion

Day 10 moved from Day 9's web-based Uncoder.IO testing to sigma-cli, the actual command-line tool used in production detection engineering pipelines. Converting the same 3 rules across 3 real SIEM backends surfaced genuine tooling limitations exactly as the task predicted — a QRadar backend unable to handle multi-condition logic, and a pipeline mismatch requiring the --without-pipeline flag. Discovering that Wazuh has no official Sigma backend, and instead relying on a community tool (SigWaz) with manual XML corrections, reinforced Day 7's lesson that real detection engineering often requires hand-fixing imperfect tooling output rather than trusting automated conversion blindly. All 5 community rules were successfully validated as live, active detections in the Wazuh ruleset.